Discrete-1.2.27

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Question

The polynomial p(x) $= x^5 + x^2 + 2$ has

a) all real roots
b) 3 real and 2 complex roots
c) 1 real and 4 complex roots
d) all complex roots

The Challenge

We need to find how many real and complex roots the polynomial $p(x) = x^5 + x^2 + 2$ has.

- Degree $= 5 \rightarrow$ Total of **5 roots**.
- Strategy: Use Descartes' Rule of Signs to narrow down the possibilities.

Explaining Descartes' Rule of Signs

This rule relates the number of **sign changes** in a polynomial's coefficients to the number of its real roots.

The Two-Step Process

1. **Positive Roots:** Count sign changes in $p(x)$.
2. **Negative Roots:** Count sign changes in $p(-x)$.

Note: The actual number of roots is either equal to the count of sign changes or less than it by an even integer.

Positive Real Roots

Write out $p(x)$ and look at the signs of the coefficients:

$$p(x) = \underbrace{+}_{\text{Term 1}} x^5 \quad \underbrace{+}_{\text{Term 2}} x^2 \quad \underbrace{+}_{\text{Term 3}} 2$$

- Sign sequence: $(+, +, +)$
- **Number of sign changes: 0**

Conclusion for Positive Roots

There are **zero** positive real roots. The graph will never touch the x-axis to the right of the y-axis.

Negative Real Roots

Substitute $(-x)$ into the equation:

$$p(-x) = (-x)^5 + (-x)^2 + 2$$

$$p(-x) = \underbrace{-}_{\text{Term 1}} x^5 \quad \underbrace{+}_{\text{Term 2}} x^2 \quad \underbrace{+}_{\text{Term 3}} 2$$

- Sign sequence: $(-, +, +)$
- **Number of sign changes: 1** (from $-x^5$ to $+x^2$)

Conclusion for Negative Roots

There is exactly **1 negative real root**.

Finding Critical Points(Cross check)

To find where the graph "turns," we take the first derivative:

$$p'(x) = 5x^4 + 2x$$

Set $p'(x) = 0$:

$$x(5x^3 + 2) = 0$$

Critical Points:

- $x_1 = 0$
- $x_2 = \sqrt[3]{-0.4} \approx -0.74$

The Second Derivative Test

We use $p''(x)$ to confirm the nature of these points:

$$p''(x) = 20x^3 + 2$$

Testing $x = 0$ (Local Minimum)

$p''(0) = 2 > 0$ (Concave Up).
Value: $p(0) = \mathbf{2}$

Testing $x \approx -0.74$ (Local Maximum)

$p''(-0.74) = -6 < 0$ (Concave Down).
Value: $p(-0.74) \approx \mathbf{2.32}$

End Behavior & Existence of Roots

We examine the limits of $p(x) = x^5 + x^2 + 2$ as x grows large:

- As $x \to \infty$, $p(x) \approx x^5 \to \infty$
- As $x \to -\infty$, $p(x) \approx x^5 \to -\infty$

Intermediate Value Theorem (IVT)

Since $p(x)$ is a continuous polynomial that changes sign from negative to positive over its domain, there must be **at least one** value c such that $p(c) = 0$.

Conclusion: This confirms the existence of at least **one real root**.

Putting it all together

- The lowest point of the "wiggle" is $y = 2$ (at $x = 0$).
- Since the local minimum is **above the x-axis**, the graph never crosses back down.
- The only time $p(x)$ is negative is when $x < -1.29$.

Conclusion: 1 real crossing, 4 hidden (complex) roots.

Python Code for Plot

```
import matplotlib.pyplot as plt
import numpy as np

# Define the polynomial function
def p(x):
    return x**5 + x**2 + 2

# Create an array of x values
x = np.linspace(-1.5, 1, 400)
y = p(x)

# Create the plot
plt.figure(figsize=(10, 6))
plt.plot(x, y, label='$p(x) = x^5 + x^2 + 2$',
    color='blue', linewidth=2)
```

Python Code for Plot

```
# Add x and y axis lines
plt.axhline(0, color='black', linewidth=1)
plt.axvline(0, color='black', linewidth=1)


from scipy.optimize import fsolve
real_root = fsolve(p, -1)[0]
plt.plot(real_root, 0, 'ro', label=f'Real Root $
    \\approx {real_root:.3f}$')

tp1_x = 0
tp1_y = p(tp1_x)
tp2_x = (-0.4)**(1/3) if (-0.4) >= 0 else -abs
    (-0.4)**(1/3)
tp2_y = p(tp2_x)
```

Python Code for Plot

```
plt.plot(tp1_x, tp1_y, 'go', label=f'Local Min
    (0, {tp1_y})')
plt.plot(tp2_x, tp2_y, 'mo', label=f'Local Max ({
    tp2_x:.2f}, {tp2_y:.2f})')

# Add labels and title
plt.title('Plot of $p(x) = x^5 + x^2 + 2$',
    fontsize=14)
plt.xlabel('x', fontsize=12)
plt.ylabel('p(x)', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.7)
plt.legend()

plt . tight_layout ()
plt . show ()
```

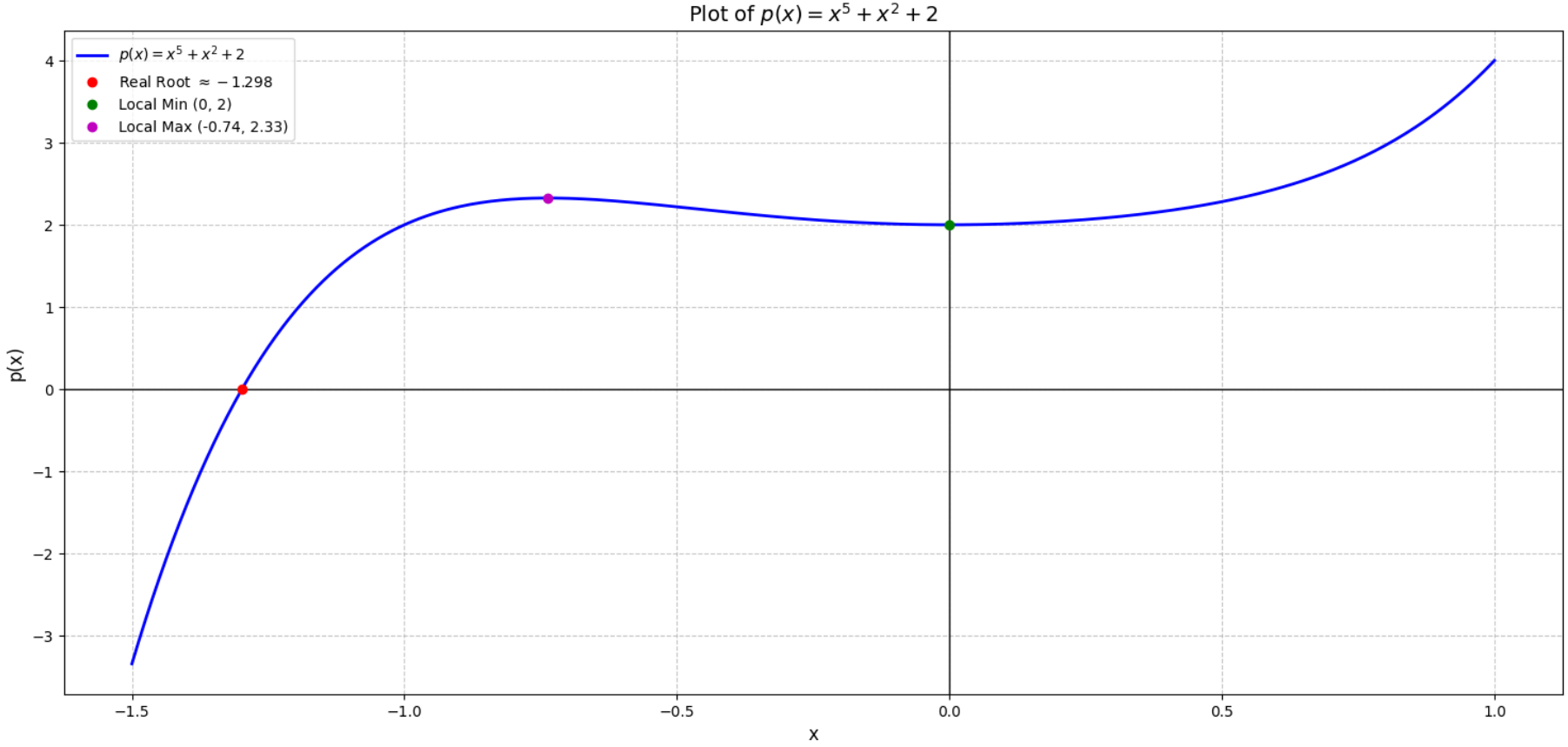

Figure: Python plot